

Generation Interconnection System Impact Study Report

For Queue Project Q22-2070

Geer 138 kV
May 02, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2070. The Interconnection Customer proposes a 385.2 MW battery energy storage facility interconnecting at the Geer 138 kV substation in the Pennsylvania area. The request was received on April 24, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 709 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 29 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$152,190,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Aec 300 - Johnsonville 138 KV Line: 232.4% loading (191 MVA) following the outage of the Vieques 2501 - Jim Bridger 138 KV line; DFAX = 100.0%.

For the loss of Jim Bridger - Vieques 2501 138 KV, the Geer - Aec 300 138 KV loads to 232.4% of its 187 MVA rating. Project DFAX: 100.0%.

The Great Works Hydro - Johnsonville 138 KV is overloaded to 214.1% of its 212 MVA emergency rating for the loss of the Vieques 2501 - Jim Bridger 138 KV L. The project contributes 100.0% DFAX (385.2 MW) to this overload.

For the loss of Procter & Gamble Mehoopany Mill - Chambes Springs 138 KV Line, the Geer - Vieques 2501 Substation 138 KV loads to 205.0% of its 219 MVA rating. Project DFAX: 100.0%.

The Great Works Hydro - Terrebonne 138 KV L is overloaded to 202.8% of its 228 MVA emergency rating for the loss of the Vieques 2501 - Jim Bridger 138 KV L. The project contributes 100.0% DFAX (385.2 MW) to this overload.

The Procter & Gamble Mehoopany Mill - Terrebonne 138 KV is overloaded to 200.6% of its 235 MVA emergency rating for the loss of the Vieques 2501 - Jim Bridger 138 KV. The project contributes 100.0% DFAX (385.2 MW) to this overload.

The Procter & Gamble Mehoopany Mill - Chambers Springs 138 KV is overloaded to 196.0% of its 246 MVA emergency rating for the loss of the Vieques 2501 Substation - Jim Bridger 138 KV L. The project contributes 100.0% DFAX (385.2 MW) to this overload.

Jim Bridger - Vieques 2501 138 KV: 189.6% loading (254 MVA) following the outage of the Procter & Gamble Mehoopany Mill - Chambers Springs 138 KV; DFAX = 100.0%.

For the loss of Strawberry Belt - Hilltop 345 KV L, the Hilltop 345/138kV Autotransformer loads to 162.3% of its 487 MVA rating. Project DFAX: 5.1%.

The Simms Substation - Wnp No. 2 Construction 138 KV is overloaded to 136.8% of its 239 MVA emergency rating for the loss of the Lakin Municipal 345/138 kV TX. The project contributes 23.5% DFAX (90.6 MW) to this overload.

The North Dayton - Grand 138 KV is overloaded to 127.4% of its 285 MVA emergency rating for the loss of the Lakin Municipal 345/138kV TX. The project contributes 23.5% DFAX (90.6 MW) to this overload.

For the loss of Lakin Municipal 345/138kV Transformer Bank, the Simms - Grand 138 KV loads to 126.6% of its 268 MVA rating. Project DFAX: 23.5%.

Wnp No. 2 Construction - Lakin Municipal 138 KV: 125.0% loading (280 MVA) following the outage of the Lakin Municipal 345/138kV Auto; DFAX = 17.2%.

The Jim Bridger - North Dayton 138 KV Line is overloaded to 121.5% of its 310 MVA emergency rating for the loss of the Lakin Municipal 345/138kV Transformer. The project contributes 23.5% DFAX (90.6 MW) to this overload.

South Hampton - Railyard Substation 138 KV: 119.6% loading (478 MVA) following the outage of the Wallenpaupack 500-345 kV Transformer; DFAX = 5.1%.

Lakin Municipal - South Hampton 138 KV: 119.3% loading (485 MVA) following the outage of the Wallenpaupack 500/345 kV TX; DFAX = 5.1%.

Lakin Municipal 345 kV/138 kV Auto: 116.9% loading (566 MVA) following the outage of the Wallenpaupack 500/345 kV TX; DFAX = 12.8%.

Borrego - Dartmouth 138 KV: 115.6% loading (204 MVA) following the outage of the Lakin Municipal 345-138 kV Transformer Bank; DFAX = 5.1%.

Johnson - Lancaster 138 KV L: 113.2% loading (185 MVA) following the outage of the Jim Bridger - Vieques 2501 138 KV Line; DFAX = 6.8%.

Kenneth Ford - Dwp 138 KV line: 111.7% loading (268 MVA) following the outage of the Geer - Vieques 2501 Substation 138 KV; DFAX = 25.3%.

Dartmouth - Jenkins 138 KV line: 109.9% loading (234 MVA) following the outage of the Lakin Municipal 345-138 kV Transformer; DFAX = 5.1%.

For the loss of Lakin Municipal 345 kV/138 kV XFMR, the Jim Bridger 138-1 kV Auto loads to 109.5% of its 420 MVA rating. Project DFAX: 45.9%.

For the loss of Lakin Municipal 345-138 kV XFMR, the Neenah Crk Substation - Borrego 138 KV loads to 107.7% of its 208 MVA rating. Project DFAX: 5.1%.

For the loss of Caldwell - Piney Crk Project 345 KV line, the Eklutna - Allegheny Hydro No 8 345 KV loads to 106.7% of its 1,188 MVA rating. Project DFAX: 5.6%.

Johnson - Lawton E Side 138 KV: 102.1% loading (196 MVA) following the outage of the Vieques 2501 - Jim Bridger 138 KV L; DFAX = 6.8%.

Woodland Tap - Seward 345 KV line: 101.5% loading (1,859 MVA) following the outage of the Allegheny Energy Units 3 4 & 5 - Southeast Vicksburg 500 KV; DFAX = 7.8%.

Weakley - Dwp 138 KV line: 101.2% loading (231 MVA) following the outage of the Kenneth Ford - Dwp 138 KV L; DFAX = 5.2%.

For the loss of Coronet - Westwoodgeneration Llc 500 KV, the Jh Waren 500 kV/345 kV Transformer loads to 101.0% of its 996 MVA rating. Project DFAX: 9.5%.

Neenah Crk - Newport Reit 138 KV line: 100.7% loading (210 MVA) following the outage of the Lakin Municipal 345 kV/138 kV XFMR; DFAX = 5.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Aec 300 - Johnsonville 138 KV	Rebuild line with higher-capacity conductor	\$6,220,000	100% (\$6,220,000)
Ae 300 - Geer 138 KV Line	Rebuild line with higher-capacity conductor	\$6,414,000	100% (\$6,414,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Great Works Hydro - Johnsonville 138 KV	Rebuild line with higher-capacity conductor	\$8,544,000	100% (\$8,544,000)
Geer - Vieques 2501 138 KV Line	Rebuild line with higher-capacity conductor	\$3,259,000	100% (\$3,259,000)
Great Works Hydro - Terrebonne 138 KV L	Rebuild line with higher-capacity conductor	\$13,115,000	100% (\$13,115,000)
Procter & Gamble Mehoopany Mill - Terrebone 138 KV	Rebuild line with higher-capacity conductor	\$10,165,000	100% (\$10,165,000)
Procter & Gamble Mehoopany Mill - Chambers Springs 138 KV	Rebuild line with higher-capacity conductor	\$844,000	100% (\$844,000)
Jim Bridger - Vieques 2501 138 KV	Rebuild line with higher-capacity conductor	\$1,744,000	100% (\$1,744,000)
Hilltop 345/138kV TX	Replace transformer with higher-rated unit	\$12,226,000	32% (\$3,972,000)
Wnp No. 2 Construction - Simms 138 KV	Rebuild line with higher-capacity conductor	\$8,886,000	96% (\$8,482,000)
N Dayton - Grand 138 KV Line	Rebuild line with higher-capacity conductor	\$2,043,000	96% (\$1,951,000)
Simms - Grand 138 KV	Rebuild line with higher-capacity conductor	\$2,685,000	96% (\$2,563,000)
Wnp No. 2 Construction - Lakin Municipal 138 KV	Rebuild line with higher-capacity conductor	\$5,469,000	100% (\$5,469,000)
Jim Bridger - North Dayton 138 KV Line	Rebuild line with higher-capacity conductor	\$3,348,000	96% (\$3,196,000)
South Hampton - Railyard Substation 138 KV	Rebuild line with higher-capacity conductor	\$4,823,000	100% (\$4,823,000)
Lakin Municipal - South Hampton 138 KV	Rebuild line with higher-capacity conductor	\$18,788,000	100% (\$18,788,000)
Lakin Municipal 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$17,977,000	100% (\$17,977,000)
Borrego - Dartmouth 138 KV	Rebuild line with higher-capacity conductor	\$3,858,000	100% (\$3,858,000)
Johnson - Lancaster 138 KV L	Rebuild line with higher-capacity conductor	\$11,704,000	83% (\$9,713,000)
Kenneth Ford - Dwp 138 KV line	Rebuild line with higher-capacity conductor	\$3,165,000	83% (\$2,626,000)
Dartmouth - Jenkins 138 KV line	Reconductor line	\$1,767,000	100% (\$1,767,000)
Jim Bridger 138-1 kV Auto	Replace transformer with higher-rated unit	\$2,792,000	100% (\$2,792,000)
Neenah Crk - Borrego 138 KV	Reconductor line	\$1,112,000	100% (\$1,112,000)
Allegheny Hydro No 8 - Eklutna 345 KV	Reconductor line	\$26,217,000	16% (\$4,291,000)
Lawton E Side - Johnson Substation 138 KV	Reconductor line	\$1,044,000	83% (\$867,000)
Woodland Tap - Seward 345 KV	Reconductor line	\$16,705,000	6% (\$919,000)
Weakley - Dwp 138 KV L	Reconductor line	\$1,564,000	58% (\$909,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jh Waren 500 kV/345 kV Transformer	Replace transformer with higher-rated unit	\$42,241,000	12% (\$4,865,000)
Neenah Crk - Newport Reit 138 KV line	Reconductor line	\$945,000	100% (\$945,000)
	Total allocated to project		\$152,190,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.